

Doc 6: Transpose, Inverse, Determinant

Module 1: Linear algebra | Part 1 of 3: Learn (theory)

1. Transpose: flip a matrix over its diagonal

You met the transpose in Doc 3: rows become columns. Now the rules that make it useful.

Definition	$(A^T)_{ij} = A_{ji}$	$[1\ 2\ 3; 4\ 5\ 6]^T = [1\ 4; 2\ 5; 3\ 6]$
Flip twice	$(A^T)^T = A$	back to the original
Sum	$(A + B)^T = A^T + B^T$	transpose each, then add
Product	$(AB)^T = B^T A^T$	the order reverses (Doc 5)
Symmetric matrix	$A^T = A$	mirror image across the diagonal, e.g. $[2\ 1; 1\ 3]$
Dot product	$a \cdot b = a^T b$	row $(1 \times n)$ times column $(n \times 1)$ gives 1×1

2. Determinant: one number that summarizes a square matrix

$$\det [a\ b; c\ d] = a\ d - b\ c$$

Example: $\det [3\ 1; 2\ 4] = 3 \times 4 - 1 \times 2 = 12 - 2 = 10$. **Meaning:** the columns of A show where the unit square's sides land. The determinant is the (signed) **area** of that parallelogram, so it is the factor by which A scales areas (volumes in 3D). **Negative** means the shape is mirrored. **Zero** means everything is squashed onto a line: information is lost, and the matrix is called **singular**.

3x3 by expanding along row 1 (signs + - +): for $[a\ b\ c; d\ e\ f; g\ h\ i]$,

$$\det = a \times \det[e\ f; h\ i] - b \times \det[d\ f; g\ i] + c \times \det[d\ e; g\ h].$$

Each 2×2 **minor** is what remains after covering the row and column of that entry.

Identity	$\det I = 1$	doing nothing keeps areas
Product	$\det(AB) = \det A \times \det B$	scalings multiply
Transpose	$\det(A^T) = \det A$	
Row swap	swapping two rows flips the sign	a mirror
Zero determinant	two equal or proportional rows, or a zero row	singular
Triangular matrix	$\det = \text{product of the diagonal entries}$	$[2\ 5; 0\ 3]$ has $\det 6$

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Part 1 of 3: Learn (theory), continued

3. Inverse: the matrix that undoes A

The inverse A^{-1} satisfies $A^{-1} A = A A^{-1} = I$ (the identity). It exists only when $\det A \neq 0$.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = 1 / (ad - bc) \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Recipe (2x2): swap the two diagonal entries, negate the two off-diagonal entries, divide everything by the determinant. Example: $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ has $\det 6 - 5 = 1$, so $A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$.

Undo the order	$(AB)^{-1} = B^{-1} A^{-1}$	like socks and shoes: undo the last step first
Transpose and inverse	$(A^T)^{-1} = (A^{-1})^T$	they commute
Inverse of the inverse	$(A^{-1})^{-1} = A$	
Diagonal matrix	invert each diagonal entry	$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/4 \end{bmatrix}$
Swap matrix	$P^{-1} = P$	swapping twice restores the order

4. Solving equations with the inverse

The system $2x + y = 5$ and $5x + 3y = 13$ is one matrix equation $A x = b$ with $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, $x = (x, y)$, $b = (5, 13)$. Multiply both sides by A^{-1} on the left: $x = A^{-1} b$.

If $\det A = 0$ there is no inverse, and the system has either no solutions or infinitely many (the equations describe parallel lines or the same line).

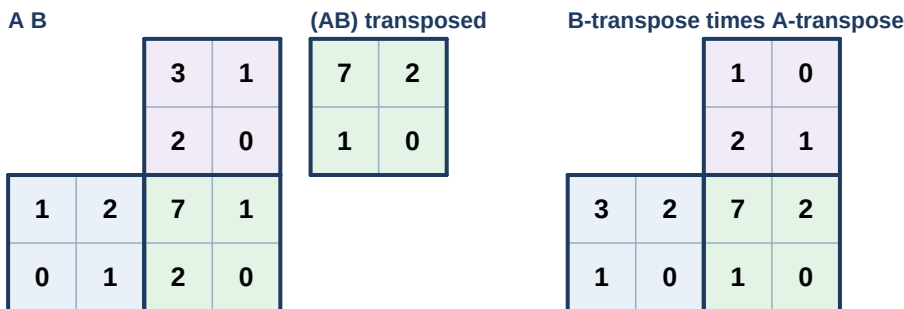
5. Where machine learning uses these

- **Linear regression** (Doc 20): the best weights are $w = (X^T X)^{-1} X^T y$. Transpose, product, inverse, all at once.
- **Covariance matrices** (Doc 9, PCA) are symmetric: equal to their own transpose.
- **Backpropagation** (Doc 24) sends gradients backward through W^T .
- $\det = 0$ means redundant features: $X^T X$ becomes singular, and regression has no unique answer.
- In practice, computers solve $A x = b$ directly instead of forming A^{-1} . It is faster and more accurate, but the inverse is the idea behind it.

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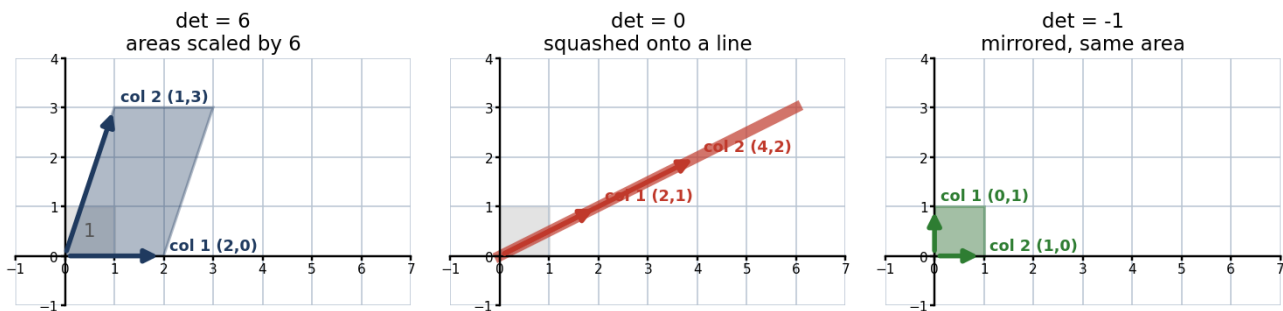
Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

Example 1: The transpose of a product reverses the order



$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix}$. Flipping AB gives $\begin{bmatrix} 7 & 2 \\ 1 & 0 \end{bmatrix}$, and $B^T A^T$ gives the same matrix. $A^T B^T$ would give something different.

Example 2: 2x2 determinants and area



$\det \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = 2 \times 3 - 1 \times 0 = 6$. $\det \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} = 2 \times 2 - 4 \times 1 = 0$ (the columns $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ point the same way). $\det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 0 \times 0 - 1 \times 1 = -1$. Grey square = the unit square, area 1.

Example 3: A 3x3 determinant by expanding along row 1

2	0	1
1	3	2
0	1	-1

Term	Sign $\times$ entry	Minor	det(minor)	Term value
Term 1	+2	$\begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix}$	-5	-10
Term 2	-0	$\begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$	-1	0
Term 3	+1	$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$	1	1

$\det A = -10 + 0 + 1 = -9$. Shortcut: a triangular matrix such as $\begin{bmatrix} 2 & 5 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & -2 \end{bmatrix}$ has $\det 2 \times 3 \times (-2) = -12$.

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Part 2 of 3: Worked examples, continued

Example 4: Finding and checking an inverse

$A = [2 \ 1; 5 \ 3]$, $\det = 1$, so $A^{-1} = [3 \ -1; -5 \ 2]$

		3	-1
		-5	2
2	1	1	0
5	3	0	1

$A = [4 \ 7; 2 \ 6]$, $\det = 24 - 14 = 10$

		0.6	-0.7
		-0.2	0.4
4	7	1	0
2	6	0	1

Each grid multiplies A by its inverse and gives the identity $[1 \ 0; 0 \ 1]$: the check that the inverse is right.
Second matrix: $A^{-1} = 1/10 \times [6 \ -7; -2 \ 4] = [0.6 \ -0.7; -0.2 \ 0.4]$.

Example 5: Solving $2x + y = 5$, $5x + 3y = 13$

		5
		13
3	-1	2
-5	2	1

$x = A^{-1} b$, with $A^{-1} = [3 \ -1; -5 \ 2]$ and $b = (5, 13)$:

$$x = 3 \times 5 + (-1) \times 13 = 15 - 13 = 2$$

$$y = (-5) \times 5 + 2 \times 13 = -25 + 26 = 1$$

Check: $2 \times 2 + 1 = 5$ and $5 \times 2 + 3 \times 1 = 13$.

Example 6: When $\det = 0$ there is no inverse

$A = [2 \ 4; 1 \ 2]$ has $\det 2 \times 2 - 4 \times 1 = 0$. Row 1 is exactly $2 \times$ row 2, so the two equations carry the same information.

System	What happens	Solutions
$2x + 4y = 6$ and $x + 2y = 3$	second equation = first $\div 2$: the same line	infinitely many
$2x + 4y = 6$ and $x + 2y = 4$	parallel lines that never meet	none

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Part 2 of 3: Worked examples, continued

Example 7: Fitting a line with the inverse (preview of Doc 20)

Data points (x, y): (2, 3), (3, 5), (4, 7). Build X with a column of 1s and a column of x-values, and y = (3, 5, 7). The best line uses $w = (X^T X)^{-1} X^T y$.

X-transpose times X				X-transpose times y				adjugate times (X-transpose y)		
			1	2						
			1	3						
			1	4						
1	1	1	3	9	1	1	1	15		
2	3	4	9	29	2	3	4	49		
									15	
									49	
								29	-9	-6
								-9	3	12

$\det(X^T X) = 3 \times 29 - 9 \times 9 = 87 - 81 = 6$, so it is invertible. The inverse is $1/6 \times [29 \ -9; -9 \ 3]$ (swap the diagonal, negate the off-diagonal). The right grid multiplies that matrix (without the $1/6$) by $X^T y = (15, 49)$ and gets $(-6, 12)$. Dividing by 6: $w = (-1, 2)$, the line $y = -1 + 2x$. Check: $x = 2$ gives 3, $x = 3$ gives 5, $x = 4$ gives 7. A perfect fit.

Example 8: $\det(AB) = \det A \times \det B$

		2	0	A = [3 1; 2 4]: $\det A = 12 - 2 = 10$ B = [2 0; 1 1]: $\det B = 2 - 0 = 2$ AB = [7 1; 8 4]: $\det(AB) = 28 - 8 = 20$ $10 \times 2 = 20$. The determinants multiply.
		1	1	
3	1	7	1	
2	4	8	4	

Quick reference card

Transpose	$(AB)^T = B^T A^T$; symmetric means $A^T = A$	
2x2 determinant	$\det [a \ b; c \ d] = ad - bc$	zero means singular
2x2 inverse	$1/(ad - bc) \times [d \ -b; -c \ a]$	check: $A A^{-1} = I$
3x3 determinant	expand along row 1: signs + - +, minors	triangular: multiply the diagonal
Solve $A x = b$	$x = A^{-1} b$	needs $\det A \neq 0$
Product rules	$(AB)^{-1} = B^{-1} A^{-1}$; $\det(AB) = \det A \times \det B$	

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Part 3 of 3: Practice. Use a pencil; yellow cells are yours. Name: _____ Date: _____

A. Transpose

1	4	2
0	-1	3

M1 (2×3)

transpose

M1 transposed

3
1
2

v (3×1)

transpose

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v transposed

2	1	0
1	3	4
0	4	5

transpose

M2 (3×3)

M2 transposed

Is M2 symmetric (M2 transposed = M2)? _____

B. (AB)-transpose equals B-transpose times A-transpose

$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$. $B^T = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$ and $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}$ are given.

A B

1	2		
3	0		

(AB) transposed

B-transpose times A-transpose

2	0		
1	3		

Are $(AB)^T$ and $B^T A^T$ equal? _____

C. 2×2 determinants

Matrix	det = ad - bc	Invertible? (Yes / No)
[3 1; 2 4]		
[5 2; 3 1]		
[2 4; 1 2]		
[1 0; 0 7]		
[0 1; 1 0]		

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Practice, continued

D. A 3×3 determinant: expand along row 1

2	1	3
0	1	4
1	0	2

Term	Sign × entry	Minor	det(minor)	Term value
Term 1	+2	[1 4; 0 2]		
Term 2	-1	[0 4; 1 2]		
Term 3	+3	[0 1; 1 0]		

det A = term 1 + term 2 + term 3 = _____

E. Inverses of 2×2 matrices

Find det A, then $A^{-1} = 1/\det \times [d \ -b; -c \ a]$. Write "none" if the inverse does not exist. Use fractions or decimals.

3	1
5	2

A

det =

inverse

A-inverse

1	2
3	4

A

det =

inverse

A-inverse

4	2
6	3

A

det =

inverse

A-inverse

F. Check an inverse by multiplying

$A = [3 \ 1; 5 \ 2]$ and $A^{-1} = [2 \ -1; -5 \ 3]$. Both products should be the identity.

A times A-inverse

		2	-1
		-5	3
3	1		
5	2		

A-inverse times A

		3	1
		5	2
2	-1		
-5	3		

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Practice, continued

G. Solve with the inverse

G1. $x + 2y = 5$ and $3x + 4y = 11$. $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Its inverse is given as the top matrix of the grid.

		5
		11
-2	1	
1.5	-0.5	

$x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$

Check both equations.

G2. $2x + y = 7$ and $x + y = 4$. $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$. First find A^{-1} , copy it into the top matrix, then multiply.

A-inverse

		7
		4

$x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$

H. Singular matrices and determinant rules

Question	Answer
1. $\det \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} = ?$	
2. Is row 1 = (2, 4) a multiple of row 2 = (1, 2)?	
3. $2x + 4y = 6$ and $x + 2y = 3$: how many solutions?	
4. $2x + 4y = 6$ and $x + 2y = 4$: how many solutions?	
5. Using A and B from exercise B: $\det A \times \det B = ?$	
6. $\det(AB)$ for the AB you found in exercise B = ?	

J. Challenge: the matrix X-transpose X (Doc 20 preview)

$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ (a column of 1s and the x-values 1, 2, 3). $X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ is given.

X-transpose times X

			1	1
			1	2
			1	3
1	1	1		
1	2	3		

$\det(X^T X) = \underline{\hspace{2cm}}$

Is $X^T X$ invertible? $\underline{\hspace{2cm}}$

Its inverse (write $1/\det \times$ the swapped matrix):

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Answer key (page 1 of 2): check your work only after finishing

A. Transpose

1	0
4	-1
2	3

 $M1^T$ | | | | |---|---|---| | 3 | 1 | 2 | |---|---|---| v^T | | | | | |---|---|---| | 2 | 1 | 0 | | 1 | 3 | 4 | | 0 | 4 | 5 | $M2^T$ |

M2 is symmetric: its transpose equals itself.

B. Product rule for transposes

A B	(AB) transposed	B-transpose times A-transpose																																				
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1	3	7	3																																			

Both give [2 6; 7 3]: equal, as the rule predicts.

C. 2x2 determinants

Matrix	det	Invertible?
[3 1; 2 4]	10	Yes
[5 2; 3 1]	-1	Yes
[2 4; 1 2]	0	No
[1 0; 0 7]	7	Yes
[0 1; 1 0]	-1	Yes

D. The 3x3 determinant

Term	Sign x entry	Minor	det(minor)	Term value
Term 1	+2	[1 4; 0 2]	2	4
Term 2	-1	[0 4; 1 2]	-4	4
Term 3	+3	[0 1; 1 0]	-1	-3

det A = 4 + 4 - 3 = 5.

E and F. Inverses

(1) det = 1, inverse [2 -1; -5 3]. (2) det = -2, inverse [-2 1; 1.5 -0.5]. (3) det = 4x3 - 2x6 = 0, so **none**.

F: A times A-inverse

		2	-1
		-5	3
3	1	1	0
5	2	0	1

A-inverse times A

		3	1
		5	2
2	-1	1	0
-5	3	0	1

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Answer key (page 2 of 2)

G. Solving with the inverse

G1

		5
		11
-2	1	1
1.5	-0.5	2

G2: A-inverse

1	-1
-1	2

G2

		7
		4
1	-1	3
-1	2	1

G1: $x = 1, y = 2$.

G2: $\det = 2 - 1 = 1, A^{-1} = [1 \ -1; -1 \ 2], x = 3, y = 1$.

Checks: $1 + 4 = 5$ and $3 + 8 = 11$. $6 + 1 = 7$ and $3 + 1 = 4$.

H. Singular matrices and determinant rules

Question	Answer
1. $\det [2 \ 4; 1 \ 2] = ?$	0
2. Is row 1 = (2, 4) a multiple of row 2 = (1, 2)?	Yes (2 ×)
3. $2x + 4y = 6$ and $x + 2y = 3$: how many solutions?	Infinitely many
4. $2x + 4y = 6$ and $x + 2y = 4$: how many solutions?	None
5. Using A and B from exercise B: $\det A \times \det B = ?$	-36
6. $\det(AB)$ for the AB you found in exercise B = ?	-36

$\det A = 1 \times 0 - 2 \times 3 = -6$, $\det B = 2 \times 3 - 1 \times 0 = 6$, and $AB = [2 \ 7; 6 \ 3]$ has $\det 6 - 42 = -36 = (-6) \times 6$.

J. X-transpose X

			1	1
			1	2
			1	3
1	1	1	3	6
1	2	3	6	14

$X^T X = [3 \ 6; 6 \ 14]$. $\det = 3 \times 14 - 6 \times 6 = 42 - 36 = 6$. Invertible: yes.

Inverse = $1/6 \times [14 \ -6; -6 \ 3]$.

Next step in Doc 20: multiply it by $X^T y$ to get the best-fit weights.

Key takeaways: $(AB)^T = B^T A^T$; $\det [a \ b; c \ d] = ad - bc$, and it measures area scaling; $\det = 0$ means squashed and singular; the 2×2 inverse swaps the diagonal, negates the off-diagonal, divides by $\det$; $x = A^{-1} b$ solves $Ax = b$; $\det(AB) = \det A \times \det B$.

Next: Doc 7 - Linear transformations and change of basis.